

Charles Darwin University
Faculty of Science and Technology
S226 PRT661 Data Science Practice

Assessment 1

Real-Time Sydney Metro Delay Prediction Using GTFS-Realtime Streaming Data and Machine Learning

Group: Sydn1- Theme1

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GitHub Link: - <https://github.com/soomansapkota/Intelligent-Traffic-Analytics>

Jira Link: - <https://suman2020j.atlassian.net/jira/software/projects/SCRUM/boards/1?filter=&groupBy=none&atlOrigin=eyJpIjoiaNzEyZGI2MzQ0ZTM3NDUxMDk1YzBkZGJkZGNjMGRkN2MiLCJwIjoiajI9>

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1. Project Overview and Objectives

Intelligent Traffic Analytics System becomes a Sydney Metro Delay Prediction System based on real-time Intelligent Traffic Analytics. It will process the TfNSW GTFS-Realtime feeds that are constantly being refreshed to create new features in combination with GTFS timetable/route feeds to make predictions of service delays 5, 10 and 15 minutes ahead. The system will concentrate on data science and predictions, rather than developing a complicated transport system, as the main aim of the project is to detect developing delays, assess the accuracy of predictions and provide valuable insights via real-time dashboard.

The main objective is:

1. To develop an end-to-end real-time data science system that continuously processes Sydney Metro operational data and predicts train delays demonstrating how real-time analytics can support faster identification of developing disruptions.

2. Theme Justification

The project directly addresses Theme 1: Real-Time Analytics and Streaming Data because the analysis is performed on continuously arriving data rather than only on a static historical dataset. The streaming layer will receive realtime transport messages, maintain short time windows and update predictions as new observations arrive. The machine-learning component adds analytical value by predicting future delay, while the streaming component demonstrates how the model can operate in a near-real-time environment.

This focus is consistent with the original proposal's emphasis on moving-stream analytics, windowed processing, continuous inference, and live visualization.

3. Problem Statement

Sydney Metro services can experience delays because of operational variation, timetable deviation, vehicle movement and service disruptions. A retrospective analysis can identify historical patterns but does not provide an estimate of what will happen several minutes into the future. The existing implementation provides a strong ingestion foundation, but a predictive layer is required to convert the realtime data into actionable information.

The research question is: Can realtime vehicle position, current delay, timetable, service-alert and temporal information be used to predict Sydney Metro service delay 5, 10 and 15 minutes ahead with useful accuracy and sufficiently low processing latency for realtime monitoring?

The project will avoid random train-test splitting because future observations must not influence earlier predictions. Instead, expanding-window or rolling-origin validation will be used, following the time-series evaluation approach specified in the original proposal.

4. Initial Architecture

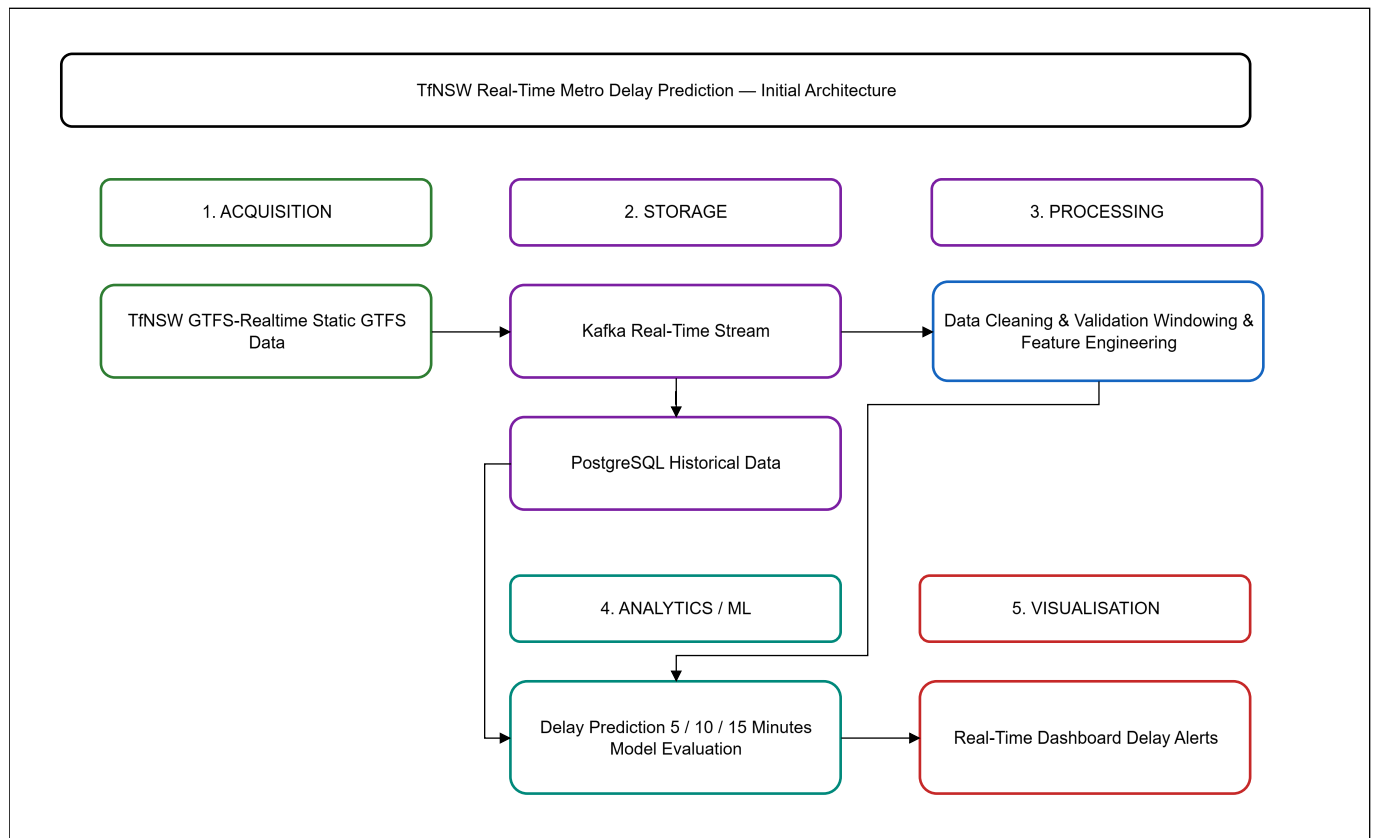


Figure 1: Proposed initial architecture for realtime Sydney Metro delay prediction.

TfNSW GTFS-Realtime feeds will be collected by a Python ingestion service and published through an MQTT/Mosquitto broker. A Python stream processor will validate incoming records and maintain one-minute windows. Features will be generated from current and historical stream state and supplied to the selected Ridge or XGBoost model. Predictions will be stored and presented through a live dashboard. The architecture remains intentionally lightweight, consistent with the original proposal's decision to avoid unnecessary Kafka and Spark infrastructure.

5. Workflow Plan

Week	Activities	Deliverable
1-2	Review existing repository; configure environment; verify TfNSW feeds and data quality.	Reliable realtime ingestion
2-3	Collect historical observations; integrate static GTFS; introduce MQTT streaming.	Streaming pipeline
3-4	Build one-minute windows; clean data; construct delay target; perform EDA.	Model-ready dataset
4-5	Engineer lag, rolling, timetable and temporal features; implement baselines.	Baseline benchmark and comparison with TfNSW prediction
5-6	Train Ridge Regression and XGBoost; initial tuning.	Candidate models
6-7	Rolling-origin validation; MAE/RMSE/MAPE; error and feature-importance analysis.	Selected model and evaluation
7-8	Deploy realtime inference; store predictions; build dashboard.	Working realtime demonstration
8	System testing; latency checks; final analysis; documentation and presentation.	Final project and report

Table 1:- Project Workflow

The eight-week plan is a compressed version of the original ten-week sprint plan. The original proposal allocated weeks 1–10 across ingestion, EDA, feature engineering, modelling and online scoring; the refined plan combines overlapping activities so the project remains achievable within the available period.

6. Task Allocation and Team Structure

Member	Role	Responsibilities
Suman Sapkota, Sachin Shrestha	Data Acquisition & Pipeline	TfNSW data ingestion, GTFS-Realtime, MQTT, streaming setup, windowing, data storage and data-quality checks
Suman Sapkota, Sachin Shrestha	Feature Engineering & EDA	Static GTFS integration, exploratory data analysis, data understanding and feature construction
Sulav Subedi, Utsav Thapa	Modelling Lead	Baseline models, Ridge Regression, hyperparameter tuning and experiment tracking
Bibek Pandit, Sulav Subedi	Model Validation & Evaluation	Model validation, performance metrics, error analysis, comparison of models and prediction reliability
Bibek Pandit, Suman Sapkota and Utsav Thapa	Dashboard & Deployment	Dashboard development, real-time prediction display, database integration, deployment and system response-time monitoring.
All Members	QA & Documentation	GitHub management, code reviews, testing, documentation, presentation preparation and final report writing

Table 2:- Team Roles & Responsibilities

Each member will take responsibility for assigned tasks while supporting and reviewing adjacent stages. All members will contribute to testing, documentation, presentation preparation and the final report.

7. Initial Risk Analysis

Risk	Likelihood	Impact	Mitigation
Insufficient historical data	Medium	High	Begin collection immediately; use available historical/static data and clearly report data limitations.
Temporal leakage	Medium	High	Use time-aware feature generation and rolling-origin validation.
Realtime feed changes or missing records	Medium	Medium	Validation checks, logging, retries and missing-data handling.
Model does not outperform baseline	Medium	Medium	Treat the result as a valid analytical finding and perform detailed error analysis.
Realtime inference is too slow	Low	Medium	Profile feature generation and model scoring; maintain incremental rolling state.
Scope creep or team availability	High	Medium	Prioritise core deliverables; keep LSTM/GRU and weather integration as optional extensions.

Table 3:- Risk Analysis

These risks are consistent with the original proposal, which identified limited history, leakage, simulated-versus-real feed differences, scoring speed and scope creep as key risks.

8. Ethics, Privacy and Security Considerations

The project analyses public transport operational information rather than passenger identities. The proposed pipeline does not require facial images, number plates or personally identifiable passenger data. Data will be aggregated at the transport-service or time-window level wherever possible.

Security controls will include protecting TfNSW credentials outside source control, using environment variables for secrets, limiting database access, validating incoming messages and keeping dependencies updated. The project will use public data according to TfNSW licensing conditions. Predictions will be presented as decision-support information and not as automated enforcement or safety-critical control.

The original proposal similarly specifies aggregate transport data, removal of personal imagery from the pipeline, protection of credentials and role-based access as core privacy and security principles.

9. References

Transport for NSW, Open Data Hub — GTFS and GTFS-Realtime datasets.

Transport for NSW, TfNSW GTFS / GTFS-Realtime Implementation Specification, June 2025.

Chen, T. and Guestrin, C. (2016). “XGBoost: A Scalable Tree Boosting System.” Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794.

Bergmeir, C. and Benítez, J. M. (2012). “On the use of cross-validation for time series predictor evaluation.” Information Sciences, 191, pp. 192–213.

Transport for NSW, Open Data Hub realtime transport and GTFS-Realtime documentation.